

# Gift A. Nwatuze — Curriculum Vitae

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## Research Interests

Software Engineering for AI Systems, Artificial Intelligence and Machine Learning, Secure and Trustworthy AI, Privacy-Preserving Machine Learning, Distributed Systems, Cybersecurity.

## Profile

Ph.D. candidate in Computer Science with expertise in software engineering, artificial intelligence, and secure distributed systems. Experienced in teaching core undergraduate and graduate courses including programming and algorithms, and in developing scalable and secure backend systems. Research focuses on building reliable and privacy-preserving AI systems for real-world applications.

## Education

- **Ph.D. in Computer Science** GPA: 3.71/4.00 — *Expected Dec. 2026*  
Kent State University, Kent, OH, USA.
- **M.Sc. in Computer Systems Engineering** GPA: 3.89/4.00 — 2019  
University of East London, London, UK.
- **B.Sc. in Software Engineering** GPA: 3.62/4.00 — 2018  
University of East London, London, UK.

## Doctoral Research Experiences

- Federated Learning for Healthcare Secure Data Sharing.
- Quantum-Resilient Federated Learning via Homomorphic-Adaptive Secure Aggregation in Decentralized Environments.
- Zero-Knowledge Proofs for Trustless Data Provenance in Multi-Modal AI Pipelines. Discover Internet of Things.
- Activity of Daily Living (ADL) Recognition using Kinect Data.
- A Framework for Integration of Machine Vision with IoT Sensing.
- Quantum-Enhanced Intrusion Detection: A Novel Hybrid Approach Using Quantum Deep Learning.
- DRNMS: An Enhanced Deep Learning-Based System for Data Recovery and Anomaly Detection in Network Monitoring.
- FairAudit: A Causal Explanation Framework for Diagnosing and Mitigating Intersectional Bias.

## Teaching and Mentoring Experiences

- **Courses:** CS 13001 (Computer Science 1: Programming and problem solving), assessment design, rubric development, feedback and mentoring, office hours.
- CS 13012 Object Oriented Programming
- CS 13011 Procedural Programming.
- CS-46101/56101-020 Design and Analysis of Algorithms.

## Adjunct Instructor — Kent State University

Jan 2026 – Present

- Independently designed and delivered undergraduate and graduate courses in Design and Analysis of Algorithms by developing comprehensive lecture materials, programming demonstrations, assessments, and course schedules aligned with departmental learning objectives.
- Applied real-world software engineering case studies, mathematical proofs, algorithm visualization tools, and live coding sessions in Python, C++, and Java to simplify complex concepts including graph algorithms, dynamic programming, greedy algorithms, and computational complexity.
- Developed assignments, programming projects, quizzes, and examinations that emphasized algorithm design, performance analysis, and practical implementation while measuring student learning outcomes.
- Evaluated student performance using structured grading rubrics, provided detailed feedback, and implemented continuous assessment strategies to improve student comprehension and academic success.
- Mentored students through weekly office hours by diagnosing programming challenges, strengthening mathematical reasoning, and guiding algorithm optimization and debugging techniques.
- Integrated interactive teaching technologies, collaborative problem-solving activities, and active-learning strategies to increase classroom engagement and participation.
- Fostered analytical thinking and software engineering best practices by connecting theoretical algorithmic concepts to practical computing applications and contemporary research problems.

## Graduate Teaching Assistant — Kent State University

Aug 2022 – Dec. 2025

- Assisted faculty in delivering undergraduate and graduate programming courses by facilitating laboratory sessions, recitations, and programming exercises.
- Supported students in learning C++, object-oriented programming, recursion, algorithm design, and computational complexity through one-on-one mentoring and office hours.
- Evaluated programming assignments, quizzes, and examinations while applying standardized grading rubrics to ensure fairness and consistency.
- Demonstrated programming concepts through hands-on coding examples and debugging sessions to strengthen students' software development skills.

- Collaborated with course instructors to prepare instructional materials, review course content, and improve student learning outcomes.

**Graduate Hourly Employee Coordinator**, Dept.of CS. Kent State University Jan 2024 - Dec 2025.

- Oversee and support student employees; ensure task efficiency and compliance with departmental standards.
- Facilitate communication between teams and leadership to improve operations.

## Professional Experience

**Software Engineering Fellow**, Coding Expert Bootcamp (Global Fellowship) Mar 2021 – Sep 2022

- Led the design of a bias-aware recommendation engine using explainable AI (XAI) and ethical constraints; results were integrated into the core curriculum.
- Developed and evaluated machine learning models using Python, implementing fairness metrics and XAI techniques to identify, explain, and mitigate algorithmic bias throughout the software development lifecycle.
- Mentored junior fellows and facilitated workshops on inclusive software design, AI ethics, and advanced software architecture.

**Software Engineer**, Trendsville / Pack Media Ltd Jan 2020 – Feb 2021

- Built secure, scalable back-end systems; integrated bias detection and XAI into internal tools; promoted DevOps best practices.

**Software Engineer**, Tech Nova Solutions May 2019 – July 2026

- Developed scalable backend systems in distributed environments.
- Designed secure APIs and optimized system performance.
- Implemented CI/CD pipelines and improved system reliability.
- Collaborated across teams to deliver production-level software solutions.

**Software Engineer**, Civil Service Commission. Jan 2016 – Dec 2019

- Automated public-sector workflows; improved reliability, security, and compliance; mentored junior developers.

## Volunteer

**Lead Software Architect & Ethical AI Advisor**, Techvision Volunteer

Jan 2021 – Dec 2022

- Directed the design and development of a mobile-first health data platform that increased rural data reliability by ~70%.
- Improved data reliability by implementing secure cloud-based data synchronization, offline data collection, automated validation mechanisms, and real-time consistency checks across distributed healthcare systems.
- Co-authored a Responsible AI policy framework adopted in emerging-economy contexts and delivered nationwide training on ethical AI and digital inclusivity.

## Doctoral Research Submission

- Nwatuze, G., Peyravi, H. *Federated Learning for Healthcare Secure Data Sharing* (Submitted to Springer Journal).
- Nwatuze, G., Peyravi, H. *FairAudit: A Causal Explanation Framework for Diagnosing and Mitigating Intersectional Bias* (Submitted to springer ai and ethics journal).
- Nwatuze, G., Peyravi, H. *Zero-Knowledge Proofs for Trustless Data Provenance in Multi-Modal AI Pipelines* (Submitted to ACM Transactions on AI Security and Privacy (TAISAP)).
- Nwatuze, G., Peyravi, H. *Quantum-Resilient Federated Learning via Homomorphic-Adaptive* (Submitted to ACM Transactions on Multimedia Computing ).

## Publications

### Doctoral Research Publications

- Nwatuze, Gift, and Hassan Peyravi. "A Framework for Integration of Machine Vision with IoT Sensing." *Sensors* 25.23 (2025): 7237.
- Nwatuze, Gift and Hassan Peyravi. "Quantum-Enhanced Intrusion Detection: A Novel Hybrid Approach Using Quantum Deep Learning. In *Proceedings of Computer Science*." *Computer Engineering and Applied Computing*, Cham: Springer Nature Switzerland, 2025.
- Nwatuze, Gift, and Hassan Peyravi. "DRNMS: An Enhanced Deep Learning-Based System for Data Recovery and Anomaly Detection in Network Monitoring." *International Conference on Computational Science and Computational Intelligence*. Cham: Springer Nature Switzerland, 2024.

### BSc and MSc Research Publications

- Nwatuze, Gift Aruchi, Comparative Performance and Optimization Strategies for Cloud-Native Architectures: A Focus on Scalability, Cost, and Resource Utilization. *International Journal of Information Engineering and Electronic Business (IJIEEB)* (2026)
- Nwatuze, Gift Aruchi, Lawrence Anebi Enyejo, and Chima Umeaku. "Enhancing cloud data security

using a hybrid encryption framework integrating AES, DES, and RC6 with file splitting and steganographic key management.” International Journal of Innovative Science and Research Technology (IJISRT) 10.1 (2025).

- Nwatuze, Gift Aruchi. ”Enhancing Security and Resilience in Cyber-Physical Systems through AI-Driven Threat Detection, Formal Verification, and Blockchain Technologies.” International Journal on Science and Technology (IJSAT) 3.1 (2025).
- Nwatuze, Gift Aruchi, et al. ”Design and evaluation of a user-centric cryptographic model leveraging hybrid algorithms for secure cloud storage and data integrity.” American Journal of Innovation in Science and Engineering (AJISE) 4.1 (2025).
- Nwatuze, Gift. ”Digital Twin Technology in Software Engineering: Simulation and Testing in Virtual Environments.” World Journal of Engineering Research 11 (2025): 322-331.
- Nwatuze, Gift. ”Human-Centered Software Engineering: Enhancing Developer Productivity and Well-Being.” World Journal of Engineering Research (WJER)11 (2025)
- Nwatuze, Gift. ”Blockchain-Based Secure Software Engineering Practices.” International Journal for Research in Applied Science and Engineering Technology (IJRASET) 13 (2025).
- Nwatuze, Gift. ”Automated Software Refactoring Utilizing Machine Learning for Extensive Systems.” International Journal for Research in Applied Science and Engineering Technology (IJRASE) 13 (2025).
- Nwatuze, G. ”Adaptive Federated Learning for Secure and Efficient Edge Intelligence in Mobile Computing” World Journal of Engineering Research and Technology (WJERT) 8.0 (2022).
- Nwatuze, G. ”Blockchain-Enabled Federated Learning for Trustworthy Multi-Agent Systems” Journal of Engineering Research and Technology (WJERT) 7.0 (2021).
- Nwatuze, G. ”Enhanced Intrusion Detection System with Machine Learning Models and Class Imbalance Optimization” Journal of Engineering Research and Technology (WJERT) 5.0 (2019).
- Aruchi, Nwatuze Gift, and Kwok Wai Jing. ”Android Device File Encryption and Decryption System.” International Journal of Information System and Engineering (IJISE) 7.0 (2019).

## Awards & Honors

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|-----------------------------------------------|-------|
| NSF Travel Award                              | 2025  |
| Certificate of Best Diplomats                 | 2025  |
| Award for Excellence in Innovation            | 2020  |
| Merit Award — Exceptional Woman in Technology | 2019  |
| STEM Outstanding Achievement Award            | 2018  |
| Dean’s Award for Academic Excellence          | 2015. |

## Certifications

- CompTIA Security certification
- Cybersecurity Threats and Defense certification
- Cybersecurity operations and controls certification
- Information Security certification

## Professional Memberships & Service

- Institute of Electrical and Electronics Engineers (IEEE)
- IEEE Women in Engineering (IEEE WIE)
- Association for Computing Machinery (ACM)
- Sigma Xi (Associate Member)
- Nigerian Institution of Professional Engineering and Scientists (NIPES) – Fellow

## Conferences & Presentations

- *Quantum-Enhanced Intrusion Detection: A Novel Hybrid Approach Using Quantum Deep Learning*, 23rd Int'l Conf. on Scientific Computing (CSC), Las Vegas, USA 2025.
- *Deep learning-based data recovery system for network monitoring (DL-DRSNM)*, Int'l Conf. on Computational Science and Computational Intelligence (CSCI), Las Vegas, USA 2024.
- *Android Device File Encryption and Decryption System*, International Conference Ascent, FMTS College, Cyberjaya Campus, Malaysia 2018.

## Peer Review & Project Reviews

**Journals:** IJIS; FCSIT; IJAIDL; IJAIML, Cogent Arts and Humanities(CAH).

**Projects Reviewed (selected):** Stock Market Data Analysis; Census Income Patterns; Movie Recommendation; NLP Language Detector; Grocery Sales Forecasting; Asset Pricing ML; Privacy-Preserving House Prices; Heart Disease Prediction; Sentiment Analysis; Customer Churn.